

MTH 446: Lecture # 17

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Lecture Span

1. Complex logs

Complex logs

Consider the analytic single-valued branch of $\log z$ as

$$\log z = \ln r + i\theta, \quad \frac{3}{4}\pi < \theta < \frac{11}{4}\pi \quad (1)$$

Recall that

$$(1 - i)^{4i} = \exp(4i \ln(1 - i)) \quad (2)$$

Then find

$$(\theta \in \arg(1 - i)) \wedge (3\pi/4 < \theta < 11\pi/4) \quad (3)$$

You have to choose the n of which $3\pi/4 < \theta < 11\pi/4$.

Analyticity of single valued of z^c

We can prove that z^c is analytic. Moreover,

$$\frac{d}{dz}(z^c) = \frac{c}{z} \exp(c \log(z)) = cz^{c-1} \quad (4)$$

Moreover, we can define the following:

$$c^z = \exp(z \log c) \quad (5)$$

Trigonometric Functions

Definition 0.1. For every $z \in \mathbb{C}$, define

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \quad \sin z = \frac{e^{iz} - e^{-iz}}{2i} \quad (6)$$

It can be proved that $\cos z$ and $\sin z$ are entire functions. Thus, $\cos, \sin \in H(\mathbb{C})$. Such functions follow the standard trigonometric differentiation rules.

You can also show

1. $\sin(z_1 + z_2) = \sin z_1 \cos z_2 + \sin z_2 \cos z_1$
2. $\sin(z_1 - z_2) = \sin z_1 \cos z_2 - \sin z_2 \cos z_1$
3. $\cos(z_1 + z_2) = \cos z_1 \cos z_2 - \sin z_1 \sin z_2$

$$4. \cos(z_1 - z_2) = \cos z_1 \cos z_2 + \sin z_1 \sin z_2$$

Prove that $f(z) = (1 - \cos 2z)/2 = \sin^2 z = g(z)$.

Proof. Since $f(z) = g(z)$ for $z \in \mathbb{R}$, by the uniqueness property, it follows that $f(z) = g(z)$ for all $z \in \mathbb{C}$. $\square$

You can also prove all the trig identities by the uniqueness property. We can also calculate

$$\sin(i) = \frac{e - \exp(-1)}{2}i \quad (7)$$

Note that $|\sin(i)| > 1$.

Also, define

$$\cosh(x) = \frac{\exp(x) + \exp(-x)}{2}, \quad \sinh(x) = \frac{\exp(x) - \exp(-x)}{2} \quad (8)$$

Define

$$\cosh(z) = \dots, \quad \sinh(z) = \dots \quad (9)$$

Note

$$1. \sin(iy) = i \sinh(y)$$

$$2. \cos(iy) = \cosh(y)$$

Calculate

$$\cos(z) = \cos(x) \cos(iy) - \sin(x) \sin(iy) \quad (10)$$

$$= \cos x \cosh y - i \sin x \sinh y \quad (11)$$

$$\cos^2 z = \cos^2 x \cosh^2 y + \sin^2 y \sin^2 x \quad (12)$$

$$= \cos^2 x (1 + \sinh^2 y) + \sinh^2 y \sin^2 x \quad (13)$$

$$= \cos^2 x + \sinh^2 y \quad (14)$$

Therefore,

$$|\cos z| = \sqrt{\cos^2 x + \sinh^2 y} \quad (15)$$

$$|\sin z| = \sqrt{\sin^2 x + \sinh^2 y} \quad (16)$$

Moreover,

$$\frac{d}{dz} \cosh z = \sinh z \quad (17)$$